No.	Status	Date
LNDE-RPT-20241202-036-DFR-1	Issued	2024-12-10

Daily Field Report 1 - PAUT Corrosion Mapping - 4-RA-101-06 and 6-RA-103-11

Scope and Responsibilities

Larkhall NDE Services Inc. was mobilized to the Hamilton Plant to perform a UT thickness survey on two rich amine lines within the ATU-100. The scope of work involves PAUT corrosion mapping of 4-RA-101 and 6-RA-103 to assess wall loss and determine remaining life based on site retirement thickness limits.

Marcus Chen, acting as the lead UT technician, is responsible for the execution of the survey, data acquisition, and ensuring all measurements are performed in accordance with the approved project specifications and safety requirements.

Regulatory and Procedural Compliance

All inspection activities were performed under QSCP-NDT-001 Rev 3. The thickness measurements and data collection methods were conducted in alignment with ASTM E797 to ensure accuracy and repeatability of the results.

Work was executed under the authority of QSCP-WP-2024-12100. Prior to the commencement of field activities, a safety review was conducted as per LNDE-JSA-20241202-036 to mitigate risks associated with the survey environment.

Equipment and Calibration

The PAUT data was acquired using an Olympus OmniScan X3 1022157, with calibration verified via LNDE-CAL-2024-0921. For spot thickness checks, Olympus 38DL PLUS 2847193 and Olympus 38DL PLUS 2901456 were utilized, with certifications LNDE-CAL-2024-0847 and LNDE-CAL-2024-0848 respectively.

Instrument calibration was performed using the LNDE-CS-001 4-step carbon steel calibration block, covering the range of 0.100-0.400 in, to ensure the equipment was properly zeroed and scaled for the A106 Gr.B material of the subject lines.

Field Execution - 4-RA-101

The survey was performed on 4-RA-101, a rich amine line with a nominal thickness of 8.56 mm and a retirement thickness of 3.81 mm. The line has a design pressure of 862 kPa(g) and a design temperature of 82 °C.

PAUT corrosion mapping was applied to the specified locations to identify any areas of localized wall loss. All readings were compared against the t-min to determine if any NCRs were required.

Field Execution - 6-RA-103

The survey continued on 6-RA-103, the rich amine header. This line features a nominal thickness of 10.97 mm and a retirement thickness of 4.57 mm, with design limits of 862 kPa(g) and 82 °C.

Thickness data was collected across the header to evaluate the current state of the A106 Gr.B piping. The mapping process was completed in accordance with the digital export requirements specified in the site procedure.

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-12-10	_____